

WASH in Schools

A complete step-by-step walkthrough of how to analyse field survey data and build a professional WASH assessment report in Microsoft Excel.

Data Cleaning · Validation · Frequency Tables · WASH Scoring Descriptive Statistics · MHM Analysis · School Report Cards

13 Sheets built	39 Schools	11 Analysis steps	55 Survey variables
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Introduction: What is WASH Data Analysis?

WASH stands for Water, Sanitation, and Hygiene. When you visit schools and collect survey data using a questionnaire, you end up with a raw dataset — a messy spreadsheet full of coded responses like 'a', 'b', 'c'. Data analysis transforms that raw data into clear, meaningful insights that guide real-world improvements in schools.

This guide walks you through every sheet in the WASH Schools Analysis workbook, explaining what each sheet does, why it matters, and how to build it yourself in Excel.

- Understand what every response code in your raw data means
- Clean messy field data and correct common data entry errors
- Build a validation log that tracks every data quality issue
- Calculate frequency tables and descriptive statistics
- Score each school using a WASH composite index (0–100)
- Identify which schools need the most urgent intervention
- Produce school-level report cards ready to share with partners
- Analyse Menstrual Health Management (MHM) as a WHO core indicator

Before You Begin: Setting Up Your Workbook

Open Excel (or Google Sheets). Create a new workbook. You will build 13 sheets, one for each part of the analysis. Name them exactly as shown below — the names make it easy to navigate and signal professionalism to any reviewer.

#	Sheet Name	Purpose
1	Raw Data	Never touch this — it is your original source of truth
2	Cleaned Data	Your corrected, standardised version of the raw data
3	Scoring Rubric	Maps every response code (a/b/c) to a point value with WHO references
4	Validation Log	Records every data error found and the action taken
5	Water Analysis	Frequency tables for all water-related questions (Q1–Q8)
6	Sanitation Analysis	Frequency tables for toilet and waste disposal questions
7	Hygiene Analysis	Frequency tables for handwashing and soap questions
8	O&M & Behavior	Operation, maintenance, Bal Sansad, supervisor accountability
9	WASH Score Ranking	Composite scores and ranking for all 39 schools
10	Descriptive Statistics	Numeric summaries: N, Mean, SD, Min, Quartiles, Max

11 Charts & Visualisations	Bar charts and pie charts for key findings
12 School Report Cards	One summary block per school — shareable with principals
13 MHM Analysis	Menstrual Health Management — WHO core indicator deep-dive

1 Understanding Your Raw Dataset

The foundation of everything. Never edit this sheet.

What the raw dataset looks like

Your raw file has one row per school and one column per survey question. The first few rows are merged headers describing section names and question numbers. Actual data starts from row 4 or 5. Most responses are letter codes: a, b, c, d — each letter maps to a specific answer choice.

Column structure at a glance

Columns	Section	Contents
A – P	Section A: Primary Info	School name, address, respondent, student counts (Boys/Girls/Special/Total), staff
Q – Z	Section B: Water	Drinking water source, quantity, storage, quality testing, rainwater harvesting
AA – AO	Section B: Sanitation	Toilet counts, gender separation, CSN access, doors, ventilation, dustbins, incinerator
AP – AW	Section B: Hygiene	Handwashing facilities, soap, MDM handwashing, height suitability
AX – BG	O&M & Behaviour	Cleaning frequency, solid/liquid waste, Bal Sansad, supervisors, MHM discussion

NOTE

GOLDEN RULE

NEVER edit the raw data sheet directly. Before doing anything, copy the entire sheet (Ctrl+A → Ctrl+C → paste into a new sheet named 'Raw Data Backup'). The raw data is your legal evidence. If you overwrite it by mistake, you lose the ability to verify your work.

How response codes work

Most questions use letter codes. For example, Q1 (Drinking Water Source):

- a = No access to drinking water at school
- b = Students bring water from home
- c = Hand pump / bore well within school premises
- d = Filtered / RO / UV / packaged water
- e = Stored rainwater (roof harvesting)

Some questions allow multiple answers. A cell might contain 'c,d' or 'b,c(Tab),d'. You handle these with wildcard formulas in your analysis (covered in Step 5).

2 Creating the Cleaned Data Sheet

Fix errors, standardise values, select the columns that matter.

The Cleaned Data sheet holds your corrected version of the raw data. You select the columns needed for analysis, correct obvious errors, and flag anything uncertain for follow-up. Do not delete raw data — just leave uncertain rows as-is and note them in the Validation Log.

Which columns to keep

#	Variable	Why include it
1	School # (Sr_No)	Unique ID — lets you trace back to raw row
2	School Name	Labels all outputs and report cards
3	Address / Village	Location context for mapping and field follow-up
4	Category	1st–4th / 5th–7th / 5th–10th — determines which norms apply
5	Shift (a=single, b=double)	Affects student load on shared facilities
6	Boys, Girls, Special, Total Students	Core enrolment data for ratios
10	Female Staff, Male Staff	Staff toilet need assessment
12	Drinking Water Source (Q1)	Multi-choice — core water access indicator
13	Water Quantity (Q2)	Conditional — only for schools with water access
14	Water Storage, Quality Tested (Q3, Q4)	Storage safety and quality assurance
16	Rainwater Harvesting (Q8)	Sustainability indicator
17	Separate Toilets (Q9)	Gender separation — SDG 6.2 indicator
18	Toilet counts Boys/Girls, Urinals Boys/Girls	Ratio-to-toilet ratio calculation
22	SN Toilets, Staff Toilets, Dustbin, Inoculator	Q12-14, 17, 18 — accessibility and MHM
26	Handwash After Toilet, Soap, Children Wash	Q16, 20, 23 — core hygiene behaviour
29	Promises Clean, Toilet Cleaning Frequency	Q26, 30 — O&M quality
31	Bal Sansad (Q33)	Student-led hygiene promotion
32	WASH Score (/100)	Calculated in Step 9
33	WASH Grade	Excellent / Good / Fair / Poor

Your first data quality check: student totals

For every school, Total should equal Boys + Girls + Special. Use this formula in a helper column:

```
=IF(F2+G2+H2=I2, "OK", "MISMATCH: "&F2;+G2+H2&" vs "&I2; )
```

Where F=Boys, G=Girls, H=Special, I=Recorded Total. Any MISMATCH goes into your Validation Log.

3 Building the Validation Log

Document every error — this is your professional audit trail.

A validation log records every data quality issue you find, what you did about it, and how serious it is. Every change to data must be documented. This protects your findings if anyone questions them — and it shows reviewers that you found the errors, not that you missed them.

Columns to set up

Column	What to record	Example
Row #	Row number in raw data	4
School Name	School name	Swami Vivekanand Vidyalay
Address	Village	Asde
Column / Field	Variable with the problem	Students_Total
Boys / Girls / Special	Raw values from each sub-column	77 / 65 / 5
Calc. Total	Your sum: Boys+Girls+Special	147
Recorded Total	What was written in Total column	142
Difference	Calc. Total minus Recorded Total	5
Severity	Minor / Moderate / Major / Flag	Moderate
Corrected?	Yes if fixed, No if flagged	No
Action / Notes	What you did or what needs to happen	Flagged for field verification

Severity classification

Minor (diff 1–2)	Likely rounding or small transcription error. Flag but do not correct.
Moderate (diff 3–10)	Possible data entry error. Flag and request original field form.
Major (diff 11+)	Critical error. Requires source document review before use.
Flag (non-numeric)	Typo in name, impossible value, missing contact. Document and correct if clear.

GOOD PRACTICE

SUMMARY ROW

At the bottom of your validation log, add: =COUNTIF(SeverityCol,"Minor") and repeat for Moderate, Major, Flag. This gives reviewers an instant data quality summary. Add the total: 'TOTAL ISSUES LOGGED: 25 | Minor: 3 | Moderate: 6 | Major: 3 | Flag: 13'

NOTE

SPECIAL CASE

When a cell has text mixed into a number (e.g. '2- slow learner' in the Special Students column), extract the number manually and note it in the Action column. Never let text in a numeric cell go unnoticed — it will break all your formulas.

4 Descriptive Statistics

Numeric summaries that describe the shape and spread of your data.

Descriptive statistics summarise numeric columns — student counts, toilet numbers, WASH scores. They tell you: what is typical (mean/median), how spread out values are (standard deviation), and the full range (min to max).

Excel formulas for each statistic

Variable / Step	Excel Formula	Notes/Result
N (count non-blank)	=COUNTA(B2:B40)	e.g. 39 schools
Mean (average)	=AVERAGE(B2:B40)	e.g. 86.3 boys
Standard Deviation	=STDEV(B2:B40)	e.g. 177.2 (spread)
Minimum	=MIN(B2:B40)	e.g. 1 student
Q1 (25th percentile)	=QUARTILE(B2:B40, 1)	e.g. 10 students
Median (50th pctile)	=MEDIAN(B2:B40)	e.g. 27 students
Q3 (75th percentile)	=QUARTILE(B2:B40, 3)	e.g. 81 students
Maximum	=MAX(B2:B40)	e.g. 1023 students

Which variables to analyse

- Students: Boys, Girls, Special students, Total enrolled
- Staff: Female, Male
- Infrastructure: Toilets (Boys), Toilets (Girls), Urinals (Boys), Urinals (Girls)
- WASH sub-scores: Water (/25), Sanitation (/30), Hygiene (/25), O&M; (/20)
- WASH composite score (/100)

INFO

SKEWED DATA

If your Standard Deviation is larger than the Mean (e.g. Boys mean=86, SD=177), your data is skewed — a few very large schools pull the mean up. In this case, always report the Median alongside the Mean. Write: 'Mean=86 (SD=177); Median=27 — data highly right-skewed due to large urban schools.'

5 Water Analysis

Frequency tables for every water question in Section B (Q1–Q8).

For each question, build a frequency table: how many schools gave each response, and what percentage that represents. Below are the exact formulas for each question type.

Single-answer questions (e.g. Q2 Water Quantity, Q4 Quality Testing)

Use COUNTIF to count each response code:

```
=COUNTIF(CleanedData!N$2:N$40, "a")
```

```
=COUNTIF(CleanedData!N$2:N$40, "b")
```

For percentage: =count_cell / N * 100 where N = total applicable schools

Multi-choice questions (Q1 Water Source) — use wildcards

Some cells contain multiple answers: 'b,c' or 'c,d' or 'c(Tab),d'. Use * wildcards to catch any cell that contains the letter:

```
=COUNTIF(CleanedData!M$2:M$40, "*a*")
```

```
=COUNTIF(CleanedData!M$2:M$40, "*b*")
```

Repeat for c, d, e. The total will be MORE than 39 schools — that is correct for multi-choice. Always add a note: 'Multiple responses allowed; total exceeds 100%'

Q2 Water Quantity — conditional question

Q2 only applies to schools that HAVE water access (Q1 answer is NOT 'a'). Schools without water skip Q2, so your N will be less than 39. Count only non-blank cells:

```
=COUNTA(CleanedData!N$2:N$40) → gives applicable N (e.g. 29)
```

Always state in your table: 'N=29 — 10 schools with no water access (Q1=a) excluded'

Sample Water Analysis table layout

Q#	Question	Code	Description	Count	% of Schools
Q1	Drinking Water Source (Multi-choice, N=39)	A	No access to drinking water	4	7.1%
		B	Students bring water from home	10	17.9%
		C	Hand pump / bore well	19	33.9%
		D	Filtered / RO / UV water	17	30.4%
		E	Stored rainwater	6	10.7%
		TOTAL	Multiple responses allowed	56	—
Q2	Water Quantity (N=29 applicable)	A	Less than 1.5 L/person/day	18	62.1%
		B	More than 1.5 L/person/day	11	37.9%
		TOTAL		29	100%

Replicate this format for all water questions (Q1 through Q8).

6 Sanitation Analysis

Toilets, CSN access, incinerators, and waste disposal.

The Sanitation Analysis uses the same frequency table structure as Water. The key questions and what to highlight from each:

Q#	Variable	Type	Key Insight to Report
Q9	Separate toilets for boys & girls	Single	% schools with gender-separated toilets (SDG 6.2 core indicator)
Q10	Number of functional toilets — Boys	Numeric	Average and range; calculate pupil-to-toilet ratio
Q11	Number of functional toilets — Girls	Numeric	Average; compare against Swachh Vidyalaya norm (1:40 girls)
Q12	Toilets for Children with Special Needs	Single	% schools complying with Disability Act — CSN accessibility
Q14	Separate staff toilets	Single	% schools where teachers share student toilets
Q15	Secure door with latch on all toilets	Single	% with full privacy — critical for girls' safety
Q16	Roof and ventilation in toilets	Single	% with adequate ventilation — reduces infection and odour
Q17	Sanitary dustbin with lid	Single	% with MHM waste infrastructure (required for girls)
Q18	Incinerator for sanitary waste	Single	SPLIT: functional vs non-functional — do not combine

Pupil-to-toilet ratio formula

A key sanitation metric is how many students share each toilet. Swachh Vidyalaya norm: 1 toilet per 40 girls, 1 per 80 boys.

```
=IF(Toilets_Girls>0, Girls/Toilets_Girls, "No toilet") → Girls ratio
```

```
=IF(Toilets_Boys>0, Boys/Toilets_Boys, "No toilet") → Boys ratio
```

Ratio below 40 for girls is good. Above 40 is a gap. Above 80 for boys is a gap.

NOTE

INCINERATOR: ALWAYS SPLIT

Raw data often has qualified 'b' responses: 'b (but not in use)' or 'b (not in working condition)'. A broken incinerator provides no MHM benefit. Report separately: functional=7, present but non-functional=3. Never merge them into one count.

7 Hygiene Analysis

Handwashing facilities, soap, and observed behaviour.

Hygiene questions capture two critical handwashing moments: after using the toilet (infection control) and before the Mid-Day Meal (food safety). You also capture whether soap is available and whether children are actually observed washing their hands.

Q#	Question	Why it matters
Q19	Handwash facility after toilet use	Prevents faecal-oral transmission — WHO core indicator
Q20	Soap provided after toilet use	Soap is the active ingredient; water alone is insufficient
Q21	Group handwash facility before MDM	Food safety — prevents contamination of mid-day meals
Q22	Soap provided before MDM	Same principle — soap must be present, not just facility
Q23	Do ALL children wash hands with soap before MDM?	Behavioural outcome — not just infrastructure existence
Q24	Height of handwashing facilities	Accessibility — suitable for all age groups in the school

Infrastructure vs behaviour gap — the most important hygiene finding

There is often a large gap between having a handwashing tap (Q19=c) and children actually using it (Q23=b). A school might have excellent infrastructure but zero behaviour change.

Calculate the gap: % with facility (Q19=b or c) MINUS % where all children wash (Q23=b)

```
=COUNTIF(Handwash_Facility, "c")/39*100 - COUNTIF(Children_Wash, "b")/39*100
```

A large positive result means children are not using available facilities — flag for behaviour change training.

8 O&M; & Behavior Change Analysis

Who maintains the school? How often? Who is accountable?

Operation and Maintenance questions cover how well schools look after their WASH infrastructure. Excellent toilets become disease vectors within weeks if nobody cleans them. This section also covers student-led hygiene promotion (Bal Sansad) and supervisor accountability.

Q#	Question	Best response	Gap response
Q25	Dustbins in all areas	b = Yes, all areas covered	a = No dustbins provided
Q26	Solid waste disposal	a = Municipality/composting	d = Open dumping (critical)
Q27	Liquid waste disposal	a = Drainage system	e = Open ground (critical)
Q28	Premises clean / no waterlogging	b = Yes, well maintained	a = Not clean
Q29	Classrooms cleaned daily	b = Yes	a = Not cleaned daily
Q30	Toilet cleaning frequency	d = Daily (recommended)	a = No schedule (critical)
Q31	Cleaning material used	c–d = Soap/disinfectant regularly	a = Water only
Q33	Bal Sansad active in hygiene	a = Yes (student participation)	b = No council active
Q34	Toilet cleaning supervisor	Named person = accountable	Vague = accountability gap

INFO

BAL SANSAD

The Bal Sansad (children's cabinet) is a student-led body that promotes hygiene in school. Schools with an active Bal Sansad tend to have better handwashing behaviour even when infrastructure is poor — because peer pressure from students works differently than teacher instructions. Highlight this in your findings.

9 WASH Composite Scoring

Turn coded responses into a 0–100 score for each school.

The WASH Composite Score is a number from 0 to 100 summarising how well each school performs across all four WASH domains. It lets you rank schools, identify priority interventions, and track improvement over time.

Domain structure

Domain	Max Points	Questions Used	What it measures
Water	/25	Q1, Q2, Q3, Q4, Q8	Access, quantity, storage quality, testing frequency
Sanitation	/30	Q9, Q12, Q14, Q15, Q16, Q17, Q18	Toilet separation, CSN access, privacy, waste disposal
Hygiene	/25	Q19, Q20, Q21, Q22, Q23	Handwashing facilities, soap, behaviour change
O&M	/20	Q28, Q29, Q30, Q31, Q33	Cleaning, maintenance, Bal Sansad
TOTAL	/100	All domains	Full WASH composite score

How to score each response (Water domain example)

In your Scoring Rubric sheet, map every response to points. Then use IFS() in Excel to convert raw answers to numbers:

Question	Response	Points	Policy Reference
Q1: Water Source	a — No access	0	JJM Guideline 1.1
Q1: Water Source	b — Bring from home	2	JJM Guideline 1.1
Q1: Water Source	c — On-site handpump	3	JJM Guideline 1.1
Q1: Water Source	d or e — Filtered/harvested	5	WHO WASH in Schools 2.3
Q2: Water Quantity	a — Less than 1.5L/person	2	JJM 15L/student/day target
Q2: Water Quantity	b — More than 1.5L/person	5	JJM 15L/student/day target
Q4: Quality Tested	a — Never tested	0	WHO drinking water quality
Q4: Quality Tested	b — Once a year	3	WHO drinking water quality
Q4: Quality Tested	c — Twice+ per year	5	WHO minimum standard

Excel IFS formula to score a single question

```
=IFS(M2="a",0, M2="b",2, M2="c",3, ISNUMBER(SEARCH("d",M2)),5, M2="e",5, TRUE,0)
```

Put scoring formulas in helper columns (hidden or in a separate tab). Then SUM the helpers for each domain.

Grade thresholds and grade formula

```
=IFS(AH2>=80,"Excellent", AH2>=60,"Good", AH2>=40,"Fair", TRUE,"Poor")
```

Where AH2 = the school's WASH composite score.

Grade	Score Range	Meaning	Recommended Action
Excellent	80–100	Meets or exceeds most WASH standards	Share as a model school; celebrate

Good	60–79	Adequate but has specific gaps	Targeted support in lowest-scoring domain
Fair	40–59	Significant WASH deficiencies	Priority intervention; budget allocation needed
Poor	0–39	Critical WASH failures	Urgent remediation; flag to district authorities

10 WASH Score Ranking

Sort all schools from best to worst with domain sub-scores.

Once every school has a WASH Score, sort them from highest to lowest. This ranking is one of the most actionable outputs — it tells decision-makers exactly which schools need the most urgent help.

Step-by-step in Excel

- Copy your Cleaned Data sheet → rename it 'WASH Score Ranking'
- Select all data → Data tab → Sort → Sort by 'WASH Score' → Largest to Smallest. Tick 'My data has headers'.
- Add a RANK column: =RANK(AH2,\$AH\$2:\$AH\$40,0) where AH = WASH Score column
- Include all four sub-scores as visible columns: Water (/25), Sanitation (/30), Hygiene (/25), O&M; (/20)
- Add conditional formatting: Home → Conditional Formatting → Highlight Cell Rules → Text Contains → 'Excellent' → Green. Repeat for Good=Blue, Fair=Orange, Poor=Red.

TIP

RANK FORMULA

=RANK(score, \$range\$, 0) gives rank 1 to the highest score (0=descending). If two schools tie, they share the same rank and the next rank is skipped. Add a secondary sort by School Name to make ties consistent.

11 MHM Analysis

WHO core indicator — analysed separately with school-level priority scores.

Menstrual Health Management (MHM) is classified by the WHO as a core WASH-in-Schools indicator. Every school with girl students must be assessed on four components: private toilet infrastructure, sanitary waste management, and MHM education frequency.

The four MHM indicators

Indicator	Raw Column	Good Response	Gap Response
Sanitary dustbin (with lid)	Q17 Sanitary_Dustbin	b = Yes, dustbin present	a = No dustbin
Functional incinerator	Q18 Incinerator	b AND confirmed functional in container, or non-functional b	
Toilet door / latch	Q15 Secure_Door	c = All toilets have secure latch	a = None present
MHM discussed with girls	Q35 MHM_Discussed	d = Monthly (WHO recommended)	a = Not discussed at all

MHM Gap Score formula (0–4 per school)

Count how many of the four indicators each school meets:

```
=IF(D2="b",1,0) + IF(I2="b",1,0) + IF(C2="c",1,0) + IF(Q2<>"a",1,0)
```

Where D=dustbin, I=incinerator, C=door latch, Q=MHM discussion. Score 4=GOOD, 0=CRITICAL.

Priority classification

Score	Priority	Action Required
4 / 4	GOOD ✓	All four MHM components in place — acknowledge and maintain
3 / 4	LOW	One gap — address in next budget cycle
2 / 4	MEDIUM	Two gaps — targeted support within 6 months
1 / 4	HIGH	Three gaps — urgent support, flag to district officer
0 / 4	CRITICAL	No MHM infrastructure — immediate intervention required

INFO

GIRL STUDENTS CONTEXT

Always cross-reference MHM gaps with the number of girl students affected. A school with score 1/4 and 923 girls enrolled is a far higher priority than one with 3 girls. Add a 'Girls enrolled' column to your MHM sheet and sort by priority then girls count.

12 School Report Cards

One summary block per school — shareable with principals and NGO partners.

A School Report Card is a condensed summary showing WASH Score, domain breakdown, and specific gaps. It is the most useful output for field workers and school principals because it is specific, readable, and directly actionable.

What each report card block should contain

- School name, address, and category
- Total students enrolled: Boys / Girls / Special
- WASH Score out of 100 and Grade (Excellent / Good / Fair / Poor)
- Domain scores: Water (/25), Sanitation (/30), Hygiene (/25), O&M; (/20)
- Percentage per domain — calculated as: $\text{=Sub_Score/Max_Score*100}$
- 1–3 specific recommendations based on the lowest-scoring domains

Sample report card (in Excel, use cell borders and background colour to mimic this):

Swami Vivekanand Vidyalay Asde WASH Score: 92/100 Grade: Excellent			
Domain	Score	Max	Percentage
Water	22	25	88%
Sanitation	25	30	83%
Hygiene	25	25	100%
O&M	20	20	100%

Repeat this block for every school, separated by 2 blank rows. All 39 schools in one sheet.

13 Executive Dashboard

The summary page — built last, links to all other sheets.

The Executive Dashboard is the first thing a donor, government official, or NGO partner will see. It summarises the entire study in one screen. Build it last — pull all numbers from other sheets using cell references so the dashboard updates automatically when data changes.

What to include

Section	What to show	Formula / Source
Study header	Title, date collected, N, district	Type manually
Key statistics	Total schools, total students, mean WASH score	=COUNTA(), =SUM(), =AVERAGE()
Grade distribution	Count of Excellent / Good / Fair / Poor	=COUNTIF(Grade_col,"Excellent") etc.
Water headline	% schools with safe water access (Q1=c,d,e)	From Water Analysis sheet
Sanitation headline	% schools with separate toilets (Q9=c,d)	From Sanitation Analysis sheet
Hygiene headline	% schools with soap at handwash (Q20=c)	From Hygiene Analysis sheet
MHM headline	% schools discussing MHM monthly (Q35=d)	From MHM Analysis sheet
Top 3 schools	Highest WASH Scores	=INDEX/MATCH from Ranking sheet
Bottom 3 schools	Lowest WASH Scores — urgent support	=INDEX/MATCH from Ranking sheet
Key narrative finding	1–2 sentence summary of the main finding	Write manually after reviewing all sheets

NOTE

ALWAYS USE CELL REFERENCES

Never type numbers directly onto the dashboard. Use =OtherSheet!Cell references. If you later fix a data error in your Cleaned Data sheet, the dashboard will update automatically. Hardcoded numbers become wrong every time data changes.

QUICK REFERENCE — KEY EXCEL FORMULAS

Variable / Step	Excel Formula	Notes/Result
Count single-choice response	=COUNTIF(Range, "a")	Exact match
Count multi-choice (any cell with 'c')	=COUNTIF(Range, "*c*")	Wildcard — catches 'b,c', 'c,d'
Percentage of total N	=Count/N*100	e.g. =C5/39*100
Check: Boys+Girls+Special=Total	=IF(B+G+S=T, "OK", "MISMATCH")	Data quality check
N of applicable responses	=COUNTA(Range)	Ignores blank cells
Score a response with IFS	=IFS(M2="a", 0, M2="b", 2, M2="c", 3, TRUE, 0)	Points from code
WASH Grade from score	=IFS(X>=80, "Excellent", X>=60, "Good", X>=40, "Average", X<40, TRUE, "Poor")	Assign grade
Domain percentage	=SubScore/MaxScore*100	e.g. =Water/25*100
MHM gap score	=IF(D="b", 1, 0)+IF(I="b", 1, 0)+...	Count of MHM met
Rank schools (highest=1)	=RANK(Score, \$ScoreRange\$, 0)	0=descending
Count schools by grade	=COUNTIF(GradeRange, "Good")	Grade frequency
Handle #DIV/0! errors	=IFERROR(formula, 0)	Shows 0 if error
Mean (average, ignores blanks)	=AVERAGE(Range)	Descriptive stat
Standard deviation	=STDEV(Range)	Spread of values
Q1/Median/Q3	=QUARTILE(Range, 1/2/3)	25th / 50th / 75th %ile
Max / Min	=MAX(Range) / =MIN(Range)	Full range
Search for text in cell	=ISNUMBER(SEARCH("d", M2))	True if 'd' found anywhere
Pupil-to-toilet ratio	=IF(Toilets>0, Students/Toilets, "No toilet")	Sanitation norm check

TIPS & BEST PRACTICES

TIP

Always save a backup first

Before cleaning data, save the raw file as WASH_data_ORIGINAL.xlsx. Never overwrite it. The raw data is your legal evidence — if you lose it, you cannot verify your work.

INFO

Name your ranges for readable formulas

Select a data column, click the Name Box (top-left corner), type a name like 'Water_Source', press Enter. Now write =COUNTIF(Water_Source,"c") instead of =COUNTIF(M2:M40,"c"). Far easier to audit.

TIP

Freeze the header row

In Cleaned Data, click on row 2. Go to View → Freeze Panes → Freeze Rows Above. Your column headers stay visible when scrolling through 39 rows.

INFO

Use IFERROR to handle blanks and zeros

Wrap any formula that might divide by zero: =IFERROR(your_formula, 0). This prevents #DIV/0! errors breaking your analysis when a school has no toilets or no data for a conditional question.

GOOD PRACTICE

Document every scoring decision

Every score you assign should be in your Scoring Rubric with a reason and a WHO/policy reference. If you score 'b' as 3 points, write why. Future analysts — or future you after 6 months — will need this to trust your methodology.

NOTE

Cross-check all frequency table totals

After each frequency table, check the TOTAL row. For single-answer questions it should equal N (39). If it does not, you have a counting error. For multi-choice, total will exceed N — confirm the note says so.

NOTE

Never confuse present with functional

A school might report having an incinerator (response b) but the Comments column says 'not in use'. Always read the comments. Separate functional from non-functional for incinerators, taps, and handwash stations.

GOOD PRACTICE

Use Claude to cross-check your analysis

Once you build your analysis, upload both the raw file and your analysis file to Claude and ask it to verify your frequency counts and statistics. It will catch mismatches between your raw data and calculations — a powerful quality check.

You now have everything you need to build a professional WASH analysis workbook from scratch. Start with Step 1 — understand your raw data — and work through each step in order. The most important habits are: never edit raw data, document every decision, and always cross-check your calculations. Well done on the analysis you have already produced — this guide will help you do it independently next time.